TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Substitution: get a variable alone, put its expression into the other equation, solve, substitute back.
Variable drops out? False statement (like 0 = 7): no solution. True statement (like 5 = 5): infinite.
Check: put both values into both equations.

1. Solve by substitution: $y = 4x$ and $2x + y = 18$.

Tip: replace y with $4x$

A. (12, 3)

B. (3, 12)

C. (2, 8)

2. You substitute and get 0 = 7. How many solutions does the system have?

Tip: false statement → no solution

F. One solution

G. Infinitely many

H. No solution

3. Write $6x + 2y = 10$ in slope-intercept form.

> **Tip:** get y alone, divide every term

A. $y = 3x + 5$

B. $y = -6x + 10$

C. $y = -3x + 5$

4. Which point is a solution of both $x + y = 10$ and $x - y = 2$?

> **Tip:** check both equations

F. (6, 4)

G. (4, 6)

H. (5, 5)